Panskura Banamali College (Autonomous)

B.Sc.(H) 3rd Semester Examination 2022 (Online mode)

Subject : Computer Science Paper: CC5P

Paper Name : Data Structure Lab

F.M. : 20 Time : 2 Hours

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2021UG009967, paper_code : CC1P , then file name will be :: 2021UG009967- CC1P.pdf for question 'A' and 2021UG009967- CC1P.mp3 for question 'B']

A. Answer any one question: 1 x12 = 12

1. Write a program to create a singly linked list having a node of at least two data fields..
2. Write a program to search for an element in a singly linked list using recursion.
3. Write a program to sort a set of nodes in a linked list.
4. Write a program of dequeue to delete an element.
5. Write a program to find trace & norm of a matrix.
6. Write a program to sort a set of strings applying insertion sort technique.
7. Write a program to apply quick sort algorithm.
8. Write a program to reverse a doubly linked list.

Marks distribution:

a. Problem Definition - 1 Marks
b. Algorithm/Flow Chart/
Pseudo-code– 5 Marks
c. Program listing - 3 Marks
d. Output/Result with discussion- 3 Marks

B. Answer any four questions: 4 x2 = 8

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What is an array ? Explain
2. What is queue? What are the various operations associated with queue?.
3. Define recursion. What are the different types of recursion.
4. Define tree. Explain with an example.
5. Define graph. State the applications of graph.

***